

Unit 5, Lesson 7: Converting Units of Length

Benchmark

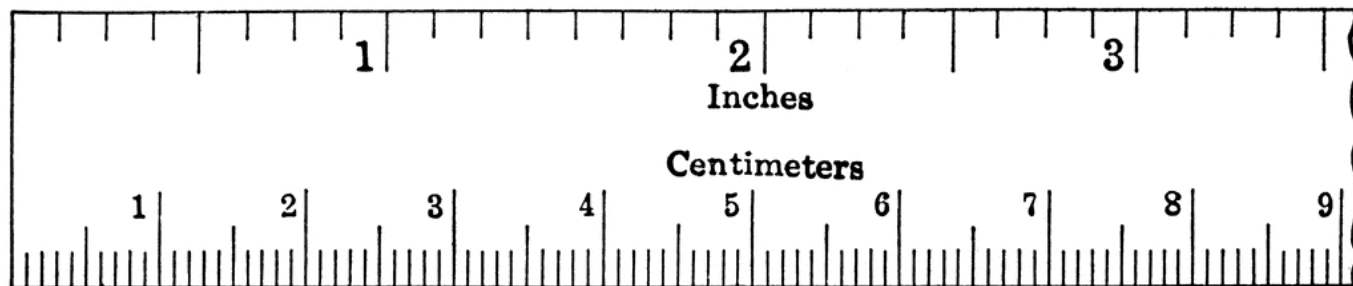
MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Learning Target

- ☐ I can convert length measurements across different measurement systems.

5.7.1 Warm-Up: A Closer Look at Rulers

A screenshot of a ruler that shows inches (in.) and centimeters (cm) is shown. The image is not to scale.



Look closely at the markings and labels on the ruler.

1. Write three observations about the markings and measurements on the ruler.
2. Compare observations with your partner. Add any additional observations your partner made to your list.

5.7.2 Activity: Measuring Objects to Determine Conversion Rates

1. With your partner, choose three classroom objects that can be measured with a standard ruler. Measure each object’s length in inches (in.) and in centimeters (cm). Use the measurements to complete the table, being as precise as possible.

Object	Length in Inches (in.)	Length in Centimeters (cm)

Similar to using exchange rates to convert currencies, you can use conversion rates to convert measurements.

To convert the length from inches (in.) to centimeters (cm), multiply the length in inches by a conversion rate. The result is the length in centimeters.

2. Use the two measurements for each object you measured to estimate a conversion rate for converting inches to centimeters.

Object 1	$\text{_____} \times r_1 = \text{_____}$	$r_1 = \text{_____}$
Object 2	$\text{_____} \times r_2 = \text{_____}$	$r_2 = \text{_____}$
Object 3	$\text{_____} \times r_3 = \text{_____}$	$r_3 = \text{_____}$

3. What is an estimate for the conversion rate of **inches to centimeters**?

$r \approx$ _____

To convert the length from centimeters to inches, multiply the length in centimeters (cm) by a conversion rate. The result is the length in inches (in.).

4. For each object, use the two measurements to estimate conversion rate for converting centimeters to inches.

Object 1	_____ $\times r_4 =$ _____	$r_4 =$ _____
Object 2	_____ $\times r_5 =$ _____	$r_5 =$ _____
Object 3	_____ $\times r_6 =$ _____	$r_6 =$ _____

5. What is an estimate for the conversion rate of **centimeters to inches**?

$r \approx$ _____

5.7.3 Bring It Together: Measurement Systems

Inches (in.) are a common unit for measuring length in the U.S. customary system, while centimeters (cm) are a common unit for measuring length in the metric system.



Did you know?

Most countries use the metric system of measurement. In fact, only three countries still use the customary system – the United States, Liberia, and Myanmar.

Conversion tables are a useful tool for converting one measurement system to another.

5.7.4 Guided Practice: Using Conversion Tables

The tables below show the conversion rates for common units of length in the US Customary and Metric systems.

U.S. Customary System to Metric System		
1 inch (in.)	=	25.4 millimeters (mm)
1 inch (in.)	=	2.54 centimeters (cm)
1 foot (ft)	=	30.48 centimeters (cm)
1 foot (ft)	=	0.305 meters (m)
1 yard (yd)	=	91.44 centimeters (cm)
1 yard (yd)	=	0.914 meters (m)
1 mile (mi)	=	1609.34 meters (m)
1 mile (mi)	=	1.61 kilometers (km)

Metric System to U.S. Customary System		
1 millimeter (mm)	=	0.039 inches (in.)
1 centimeter (cm)	=	0.39 inches (in.)
1 centimeter (cm)	=	0.033 feet (ft)
1 meter (m)	=	39.37 inches (in.)
1 meter (m)	=	3.28 feet (ft)
1 meter (m)	=	1.094 yards (yd)
1 kilometer (km)	=	1093.613 yards (yd)
1 kilometer (km)	=	0.62 miles (mi)

Complete the statements below using the information from the conversion tables.

1. To convert feet to centimeters, multiply the number of feet by _____ to find the equivalent number of centimeters.
2. To convert centimeters to feet, multiply the number of centimeters by _____ to find the equivalent number of feet.

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3. To convert meters to yards, multiply the number of meters by _____ to find the equivalent number of yards.
 4. To convert yards to meters, multiply the number of yards by _____ to find the equivalent number of meters.
 5. Use the conversion rates to convert each measurement length.
 - a. 30 miles $\approx$ _____ kilometers
 - b. 65 kilometers $\approx$ _____ miles
 - c. 10 inches $\approx$ _____ millimeters
 - d. 450 millimeters $\approx$ _____ inches

5.7.5 Collaboration: Digging Deeper into Conversion Rates

The calculations below show how 5 feet (ft) can be converted to centimeters.

$$\frac{5 \text{ ft}}{1} \times \frac{30.48 \text{ cm}}{1 \text{ ft}} = 152.4 \text{ cm}$$

Koji said the factor $\frac{30.48 \text{ cm}}{1 \text{ ft}}$ is equivalent to 1 because the numerator and denominator are equivalent values.

1. With your partner, discuss what Koji used to say that $\frac{30.48 \text{ cm}}{1 \text{ ft}}$ is equivalent to 1.
2. Explain how this shows 5 ft is equal to 152.4 cm.
3. Eva noticed that in the calculation of the conversion, the unit feet (ft) was in the numerator and the denominator. How does this eliminate the units of feet in the calculation?

4. Complete the statements below.

When multiplying a measurement by a conversion factor, the value of the

measurement

- ☐ does
- ☐ does not

change because the conversion factor is equivalent

to 1.

The units

- ☐ do
- ☐ do not

change because when the same units appear in the

numerator and denominator, it is similar to having the same value in the

numerator and denominator, so these units become 1.

***Multiplicative Identity Property of One***

$$a \times 1 = a \text{ or } 1 \times a = a$$

5.7.6 Try It: Converting Lengths to Solve Problems

1. Raphael measures the height of a plant each week using an inch (in.) ruler. On week 5, he cannot find the inch ruler, so he uses a centimeter (cm) ruler instead. His measurements are shown.

Plant Height

Week	1	2	3	4	5	6
Height	3.4 in.	4.2 in.	5.0 in.	5.8 in.	16.8 cm	7.4 in.

How much taller, in inches, is the plant in week 5 than in week 4?

5.7.7 Your Turn!

1. Use the conversion rates to convert between the lengths below. Round to the nearest thousandth when necessary.
 - a. 25 miles $\approx$ _____ kilometers
 - b. 25 centimeters $\approx$ _____ feet
 - c. 3 yards $\approx$ _____ meters
 - d. 470 millimeters $\approx$ _____ inches
2. Maya visits her friend in Europe and rides in an electric car. The electric vehicle can travel 200 kilometers (km) on one charge. If their destination is 130 miles (mi) away, will they be able to make it there without having to stop to charge the car? Explain or show work to justify your answer.

5.7.8 Wrap-Up: Ticket Out the Door

Darnell and Jada both walk every day for exercise. Darnell averages 6.8 kilometers (km) per day while Jada averages 4.5 miles (mi) each day. Darnell claims he walks farther than Jada, but Jada disagrees.

1. Without performing any calculations, which student do you think is correct? Explain your reasoning.
2. Use unit conversions to determine who walks farther each day.